
Pattern Arithmetic: Typed Algebra of Bounded Units on a Phase-Colored Sphere

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September 17, 2026

Abstract

Ordinary arithmetic treats the integer 1 as a unit of truth: $1 = 1$ and $1 + 1 = 2$. Counting is only one algebra. Colors and musical notes are also units of truth; combining them yields a hue or a chord, not a larger integer. This paper treats a pattern as a higher-order unit and represents one pattern by one typed point $x = (r, \hat{n}, \chi, \tau)$ on a unit sphere $\mathcal{S} \subset \mathbb{R}^3$: strength $r \in [0, 1]$, direction $\hat{n} \in S^2$ (equivalently (θ, ϕ)), internal phase-color $\chi \in [0, 2\pi)$, and sort τ . Pattern Arithmetic is the algebra of such units. It proceeds in three acts — collapse (the bowl, $\oplus$), transform (the path, T), result — so that gather does not double identical states ($A \oplus A = A$), a gathered unit may be removed ($\{A\} \ominus A = \emptyset$, the empty bowl), and the same ingredients may form different ordered expressions. Anti-phase cancellation to black is a Mix reading, not a gather. Division as unmixing is left open. Words and numbers occupy the same sphere under different encodings and meet in a document by order and typed action, not by blending. The aim is a single bounded register that carries overlapping identities, structural paths, and mixed prose without a discrete grid. The sphere and its operators are one proposed encoding of a narrower claim — a combination rule should match the unit — not a theory of language, physics, biology, or computation.

1 Introduction

Ordinary arithmetic uses the integer 1 as a unit of truth. The identity $1 = 1$ says that two tokens are of the same kind. The rule $1 + 1 = 2$ says that combining two such tokens produces a larger quantitative state. That algebra answers the question *how many?*

It is not the only algebra. A color and a musical note are also units of truth, but their combination does not enlarge a pile. Under additive light, red and green yield yellow. Two copies of the same yellow, once brightness is normalized, remain yellow. Idempotence is a

property of $\oplus$ alone: $A \oplus A = A$ always, whereas $A \boxplus A$ grows strength to $r' = \min(1, 2r)$ at unchanged hue, and returns the same yellow only after that normalization (Axiom 2.7). Two notes yield a chord, not a larger integer; a tone plus its opposite phase yields silence. In each case the word “add” names a different rule, and therefore a different kind of unit.

When the object of interest is a structured whole rather than a count, a hue, or a pitch, the natural unit is a *pattern*. This paper represents one pattern by one typed point on a unit sphere $\mathcal{S} \subset \mathbb{R}^3$, not by a field painted over the globe:

- $r \in [0, 1]$ is the strength of the unit, capped so that a state remains a single unit of truth — an activation, not a global probability density;
- $\hat{n} \in S^2$, or equivalently (θ, ϕ) , is the direction of the pattern on the sphere;
- $\chi \in [0, 2\pi)$ is an internal phase-color, so two identities may occupy the same direction without overwriting one another;
- τ is the sort (**word**, **num**, **color**, ...): a legality tag, not a fifth geometric angle.

Pattern Arithmetic is the algebra of these units. Gather does not double identical states ($A \oplus A = A$). A gathered unit may be removed ($\{A\} \ominus A = \emptyset$). Rotations of the sphere preserve the unit. Opposite-phase Mix is a later reading; it is not $\oplus$.

A single numerical picture is enough to see the difference from counting. Take two unit states at the same place, one with phase 0° (red) and one with phase 120° (green). Counting would return 2. Phase-aware combination returns a single unit whose blended phase is 60° (yellow). That is this paper’s color inclusion — phasor Mix on a toy chart, not CIE or linear RGB (Axiom 2.1, Axiom 2.7). The same two strengths with opposite phase (0° and 180°) return the null state, black. Two units of truth may become a new identity, a stronger mark of the same identity, or none.

The working order of the paper is not that mix alone. It is a three-step spine:

$$\text{Collapse (bowl)} \longrightarrow T \text{ (path)} \longrightarrow \text{Result}.$$

Collecting red and green is not yet yellow, and not yet a poem. Yellow is what you get if the later reading is Mix ($\boxplus$). Red and green may equally well walk a T_{poem} path; the result need not be yellow — though a given poem may still arrive there. The later act, not the shells, chooses the result. A verse is another path. Each word of “Twinkle, twinkle, little star...” is a unit state on the same sphere. Collapse yields the bag of distinct words (**twinkle** appears twice in the line and only once in the bowl). The poem is the path that walks those states in order, including the repeated **twinkle**. The bag is not the song. Numericity is not stored by $\oplus$. Two identical units collapse to one address; that is the point of $A \oplus A = A$. How many times the address is *visited* lives in the path (the tape says **twinkle** twice). How many as a *count* lives in **num** mode, where $1+1+1+1=4$ and four tokens of 1 are not collapsed. Strength r is a poor substitute for either: a second **twinkle** cannot honestly raise r past 1, and “four ones” is schoolbook addition, not a fuller unit of truth. The same bowl may be read more than one way. Loading is not meaning.

The sphere is the stage. It is not the deepest claim. Strip away geometry, phase, rotations, and the language examples, and one proposition remains: different kinds of information require different notions of combination, and an operation should be read by what informational property it *preserves* — and what it is allowed to throw away.

$$1 + 1 = 2, \quad \text{red} \boxplus \text{green} = \text{yellow}, \quad A \oplus A = A, \quad T_a(T_b(x)) \neq T_b(T_a(x)).$$

These are not alternative tricks for the same plus sign. They answer different questions:

Operation	Question
$+$	How many?
$\oplus / \ominus$	What is present? / what is removed?
$\boxplus$ (Mix)	What composite state results?
T	What happens, and in what order?
offset	What walk is reused?

Division as deconvolution or signal unmixing is not among these verbs; it is left open (Section 7.2). Inverse rotation of T is well-defined. A Mix inverse is not.

A pattern can be treated as one unit. Identity and multiplicity come apart: $\oplus$ keeps kind and discards count; the path keeps visits; **num** keeps quantity. Composition and transformation are different verbs. Sorts may share a stage without sharing an operator. The geometry that follows is only a language in which those verbs can be written. It does not mint the relations; a lexicon or a learning rule must still seat Sun and Day. Section 1.4 keeps that claim narrow.

1.1 The Limitations of Discretized Matrix Grids

A discrete matrix grid is the wrong box for such a unit. Traditional matrix representations of continuous state spaces suffer from two severe structural bottlenecks:

- **Memory Scaling Bottlenecks:** A voxel cube of resolution N stores $O(N^3)$ cells, growing steeply with resolution.
- **Signal Identity Destruction:** Mapping multiple overlapping variables to identical spatial coordinates often results in the destructive overwriting of underlying signal identities.

1.2 The Phase-Colored Unit Sphere Framework

To resolve these grid constraints, the sphere is offered as one bounded object that can carry overlapping identities by direction and phase, without enlarging a lattice.

- **Spatial Orientation** $\hat{n} \in S^2$, or (θ, ϕ) : Defines continuous directional coordinates across the spherical surface.
- **Magnitude** ($0 \leq r \leq 1$): Strength of that one unit, capped so the arrow remains a single unit of truth. An activation, not a density for the whole sphere.
- **Internal Color Phase** ($\chi \in [0, 2\pi)$): Serves as an independent phase degree of freedom.
- **Sort** τ : A legality tag (**word**, **num**, **color**, ...). It does not move the arrow.

1.3 Core Contributions & High-Level Intuition

The principal contributions of Pattern Arithmetic are:

- **A typed unit:** one atom $x = (r, \hat{n}, \chi, \tau)$ on a bounded sphere, not a voxel cell and not a global probability field.
- **Admissibility of sorts (Axiom 0):** a sort earns a geometric χ - or T -reading only where a stated map preserves its native combination; most domains do not.
- **Verbs on that unit:** $\oplus$ collects identities ($A \oplus A = A$); $\boxplus$ reads a composite; T walks an ordered path on one register occupant; offset copies a relation. Schoolbook $+$ is reserved for `num`.

Later sections distinguish a numeric encoding from a lexical encoding on the same sphere, so that a report may carry words and numbers together without forcing them through $\oplus$.

1.4 Scope of the claim

The actual claim is narrow. Counting ($1 + 1 = 2$) is not the only valid arithmetic for combining units of information. A combination rule should be chosen for what the unit *is* — a count, a color, a direction, a word — not applied by default because it is the arithmetic everyone already knows. The sphere, phase, and the operators $\oplus$, $\boxplus$, T , and offset are one proposed way to make that claim precise and checkable. They are not themselves the claim, and they are not offered as a general theory of language, physics, biology, or computation. The paper does not attempt to compute the governing equations of those systems. It attempts to arrive at higher-order realities by reading many co-present units of truth — seated units, whether pinned by a lexicon or by a learned map — through a path. That is a computation of *presence, phase, and order*, not a substitute for a field equation. Storms, forests, a table and chair, and a pathogen cancellation later in the paper (Section 8) are costumes for the $\oplus$ -versus- T split: presence versus path. Related-work paragraphs (Section 9) draw boundaries against existing formalisms. None of that is a demonstrated model of weather, ecology, or epidemiology, or a replacement for quantum mechanics, embeddings, or transformers. A later widening of a costume into a domain claim would need its own evidence. It is not an extension of the illustration. Section 2.1 states this limit formally: a sort earns a geometric reading only where a structure-preserving map into Mix or T exists at a stated fidelity, and most physical domains do not offer one.

The next section turns these definitions into formal mathematical axioms.

2 Axiomatic Foundations Of Pattern Arithmetic

2.1 Axiom 0: Admissibility of Sorts

The axioms that follow govern how a state $x = (r, \hat{n}, \chi, \tau)$ behaves once it sits on $\mathcal{S}$. They say nothing about which sorts τ have earned the right to sit there with a geometric reading attached. That prior question is this axiom, and it is separate from Table 2 in Section 5.4: the table says which *operators* two already-admitted sorts may use on each other; this axiom says which sorts may claim χ and T mean something about their own domain in the first place.

The test. A sort τ has a domain D_τ and, in the world, its own native way of combining two members of that domain, written $a \circ_\tau b$ for $a, b \in D_\tau$. Call τ *geometrically admissible* if there is a map

$$\varphi_\tau : D_\tau \longrightarrow \mathcal{S}, \quad \varphi_\tau(a \circ_\tau b) \approx \boxplus(\varphi_\tau(a), \varphi_\tau(b)) \quad \text{or} \quad T_{\varphi_\tau(a)}(\varphi_\tau(b)), \quad (1)$$

holding at a stated fidelity (exact, or approximate with the approximation named). Equation (1) is the Tier 1 test. Tier 0 admission, below, does not require it. If no such φ_τ exists, τ is *geometrically inadmissible* for Tier 1: it may still occupy a point on $\mathcal{S}$, but only under the inert treatment described below, never with a claim that χ or T tracks anything the domain itself does.

Three tiers.

- **Tier 0 — default, unstated.** `word` and `num` need no admissibility argument because neither makes a physical claim through the geometry. `num` has an exact, invertible *chart* (Section 5.2): decode, add in $\mathbb{Z}$, encode. That chart is not a Mix or T homomorphism — Mix on it is the clock trap of that same section, which is why Mix on `(num, num)` is type-refused. `word` makes no claim at all — T on a word is bookkeeping (order, presence), never a statement about linguistic physics. Mix applied to two word-states is inert geometry, not a χ -reading of language. Both are silently admitted at this inert level; nothing further needs to be said to use them.
- **Tier 1 — explicit inclusion.** A sort earns a genuine χ -reading only by stating its φ_τ and the fidelity it holds at. Musical pitch class is close to exact: octave equivalence makes pitch native to a circle, so χ for a note is close to literal, not borrowed. Color is close but not unique: additive light mixing is linear in some models and only approximately phasor-like in others, so an inclusion of `color` must say *which* reading it means — Mix under $\boxplus$ and Mix under linear RGB blend are different, non-interchangeable inclusions, and a paper that admits color must pick one and say so. This paper picks phasor Mix on a toy $0^\circ/120^\circ/60^\circ$ chart (Axiom 2.7). It does not pick CIE, and it does not pick linear RGB. A second Tier-1 sort, `chladni`, is worked in full in Section 2.8, where Equation (1) is met exactly within a mode and shown unavailable across modes.
- **Inadmissible — no φ_τ exists.** Some domains have a native combination law with no structure-preserving map to $(\oplus, \boxplus, T, \text{offset})$ at any useful fidelity. A snowflake’s native symmetry is the discrete point group D_{6h} , and two ice structures meeting is a diffusion-limited accretion process governed by temperature and humidity gradients — not a continuous rotation in $\text{SO}(3)$. There is no honest $\varphi_{\text{snowflake}}$ satisfying Equation (1). Such a sort is not barred from the register outright: it may still be loaded as a point, but only demoted to the Tier-0 treatment already given to `word` — T used for order or presence only, no χ -reading granted, and no claim that a rotation of its point corresponds to anything the domain does. What is barred is Tier-1 status: an inadmissible sort may never be handed a stated φ_τ , because none would be true.

This axiom is also what gives Section 1.4 teeth. “Not a general theory of physics, biology, or computation” is not only a promise of restraint; it is a consequence of Equation (1) failing for most physical domains a reader might otherwise expect the sphere to cover.

Sort	Native law $\circ_\tau$	Mix/ T map?	Tier
num	schoolbook +	no (chart only)	0
word	none claimed	n/a	0
note-pitch	pitch-class sum	near-exact	1
color	additive light	phasor Mix (here)	1
chladni	modal superposition	exact, within a mode	1 [§]
snowflake	D_{6h} growth	no	T1 no; T0 yes

Table 1: Admissibility is a property of the sort, decided before Table 2’s operator legality is even asked. Equation (1) is the Tier 1 test; Tier 0 admission does not use it. A sort absent from this table is undecided, not admitted; treat it as inadmissible until a φ_τ is stated. [§]chladni is admitted at Tier 1 with a stated restriction: exact for superposition within one mode, undefined across modes (Section 2.8).

2.2 Process principle: Collapse, path, result

Pattern Arithmetic proceeds in three named acts, kept separate so that a bowl may grow before it is read:

$$\underbrace{A_1 \oplus A_2 \oplus \cdots \oplus A_n}_{R = \text{the bowl}} \longrightarrow T_{w_k} \circ \cdots \circ T_{w_1}(x) \longrightarrow \text{Result}. \quad (2)$$

Here x is the register occupant; the w_i are actors, taken from the bowl or from the tape. T does not act on the set R as a set. $\oplus$ *collects*: identical tokens do not double ($A \oplus A = A$), and opposite marks may sit together until a later reading. T is an ordered path, not a mix. The result is whatever that path (or another named reading) makes of the bowl — a poem, a report, a staged endpoint.

Paint is Mix ($\boxplus$), a reading of peer states at a common direction, not a T :

$$\text{red } \boxplus \text{ green} = \text{yellow}.$$

The same pair may take T_{poem} instead. That result is a verse about the two colors, not a hue — unless the poem itself walks back to yellow. Shells do not own a pathway. A verse is another path. One bowl, several ways to read it. One may collect and read in a single gesture; the algebra does not force the merge. This principle is an axiom of *order of acts*, not a new formula for a 4-tuple.

Levels are closed. A result may be seated again. Equation (2) returns a state, and a state may take a seat in a further bowl, so the three acts repeat at a higher level: words gather and are walked into a line; the line is seated; lines gather and are walked into a verse. Two properties hold at every level and are worth stating once. First, a unit carries no representation of what it constitutes. A seated word does not know the poem, and T leaves τ_x invariant (Section 3.3), so being acted upon does not change what a unit is. Second, a bowl carries no representation of the reading that will be applied to it; that is what *one bowl, several ways to read it* means above.

The consequence is a design rule rather than a theorem. Whatever a lower unit would have to know about a higher one belongs at the higher level and not in the 4-tuple: which units were taken as co-present, over what interval, within what region, and under what question. Those are constituting facts of a level, not coordinates of an atom, and keeping them out is why the atom stays at four coordinates and one sort tag while the levels

above it may be as elaborate as a problem requires. This paper works a single level and states the mechanism for the next; it does not carry a multi-level example.

2.3 Axiom 1: The Normalized Unit Truth State ($r \leq 1$)

A pattern is one typed point, not a field painted over the globe. The working atom is

$$x = (r, \hat{n}, \chi, \tau), \quad \begin{array}{ll} r \in [0, 1], & \hat{n} \in S^2, \\ \chi \in [0, 2\pi), & \tau \in \{\text{word, num, color, note-pitch, chladni, } \dots\}. \end{array} \quad (3)$$

Equivalently, $\hat{n}$ may be written in polar angles (θ, ϕ) , so the geometric part of the atom is the 4-tuple (r, θ, ϕ, χ) used by the operators below. The sort τ does not move the arrow; it says which buttons may be pressed (Section 5). A collection (bowl) is a separate finite set $B = \{x_1, \dots, x_n\}$ of such atoms. The sphere holds one current occupant of the register; the bowl is who is present.

The cap $r \leq 1$ applies to *that* arrow. It is a normalized strength / activation coordinate, not a global probability density and not a demand that all arrows on $\mathcal{S}$ integrate to one. Two states may occupy different directions at once; each still carries its own strength $r \leq 1$. The sphere is the stage. The unit is the vector. $r = 1$ is a full-strength mark. Nothing in this axiom sums the globe into a single global budget. A field of many atoms, if wanted later, is a different object (for example a superposition of Dirac masses at those points), not the atom itself.

How the bound is imposed is a choice. This paper bounds by clipping: Mix returns $r' = \min(1, |z'|)$, so $r = 1$ is attained whenever a phasor sum reaches unit length. Clipping is not injective. Every sum of length one or more returns the same r' , and agreement past that point is not recorded (Section 2.7). An order-preserving bijection from $[0, \infty)$ onto $[0, 1) — |z| \mapsto |z|/(1 + |z|)$, say — would bound without discarding, since distinct sums would keep distinct strengths, and it would leave $z = 0 \mapsto 0$ and the black of Axiom 2.4 untouched.

The clip is kept here, and the trade is stated rather than argued. Under such a bijection $r = 1$ becomes a limit and not a value: the saturated pole of Axiom 2.5 would be approached and never occupied, and $\alpha = \pi r_w$ would never reach the half-turn of Section 3.3. Two attained poles are worth more to this paper than lossless accumulation, because no construction here sums a large bowl. A setting that does — a verdict accumulated from many units, say — would weigh the same trade the other way, and should revisit this line rather than inherit it.

2.4 Axiom 2: The Absolute Null State ($r = 0$, Black)

The lower bound of a single unit is vanishing strength:

$$r = 0 \quad \implies \quad x = \mathbf{0} \quad (\text{Black}). \quad (4)$$

Direction and phase are then unused: there is no arrow to point. Black is an empty seat, not a field that is zero at every angle.

2.5 Axiom 3: Saturation (White)

White is the upper pole of the same bounded stage. It is not an infinite norm. Two readings are allowed; neither leaves the cap $r \leq 1$.

White as a colour. A single arrow may take a full-spectrum name (white light as one identity). That state is still one 4-tuple with $r \leq 1$.

White as a full house. The bowl may be saturated: many directions occupied, many hues present. Collecting further distinct shells fills more seats; it does not raise any arrow past $r = 1$. $\oplus$ does not grow strength. Mix may grow a single r until the cap. The stage looks white because the palette is on. The catalogue of names is large — on the order of a 24-bit palette, ~ 16 million colours per direction — and may be called practically endless. No unit is $r = \infty$.

$$\text{White} = \text{saturation of the bowl, or a full-spectrum name.} \quad (5)$$

Black is empty. White is as full as this construction allows. The menu is huge. The arrows stay finite. Density in this paper means many identities can share the stage without overwrite — many directions, and many phases per direction — each still capped at $r \leq 1$. It does not mean infinite information, and it is not a storage theorem against a voxel grid.

2.6 Axiom 4: Gather and remove ($\oplus$, $\ominus$)

$\oplus$ *collects* a unit into the bowl. It does not mix colours and it does not turn an arrow. A bowl is a finite subset of $\mathcal{S}$, and $\oplus$ is union on such subsets:

$$\oplus : 2^{\mathcal{S}} \times 2^{\mathcal{S}} \longrightarrow 2^{\mathcal{S}}, \quad B_1 \oplus B_2 = B_1 \cup B_2.$$

A bare state A appearing in an $\oplus$ -expression denotes its singleton bowl $\{A\}$. With that convention, identical names occupy one seat:

$$A \oplus A = A, \quad \text{that is,} \quad \{A\} \cup \{A\} = \{A\}, \quad (6)$$

while distinct names take separate seats, $A \oplus B = \{A, B\}$, so opposite marks may sit together: $\{+1\} \oplus \{-1\} = \{+1, -1\}$. Idempotence, commutativity and associativity of $\oplus$ are those of union, and none of the three needs separate postulation. Gather is sort-blind: any two seated units may share a bowl, whatever their sorts, since a bowl records presence and presence makes no claim about a domain (Table 2). Sort discipline governs what may later be *done* with the bowl, not who may sit in it. The bowl is a set of states, not a blended 4-tuple.

$\ominus$ *removes* a unit, $\ominus : 2^{\mathcal{S}} \times \mathcal{S} \rightarrow 2^{\mathcal{S}}$. Removing the last occupant leaves the empty bowl:

$$\{A\} \ominus A = \emptyset. \quad (7)$$

The empty bowl $\emptyset$ and the null state $\mathbf{0}$ of Axiom 2.4 are distinct objects: $\emptyset$ is a bowl with no occupant, $\mathbf{0}$ is an occupant with $r = 0$. Only $\mathbf{0}$ is a state, which is why $T_{\mathbf{0}}$ is defined (Section 3.3) and $T_{\emptyset}$ is not. $\ominus$ undoes a gather. It is not anti-phase interference. Idempotence then gives the counting-trap

$$A \oplus A \oplus A \oplus A \ominus A = \emptyset \quad (\text{the empty bowl}). \quad (8)$$

Four gathers of the same name are still one seat; one removal empties it. Schoolbook $1 + 1 + 1 + 1 - 1 = 3$ lives only in **num** mode.

2.7 Axiom 5: Mix as a reading ($\boxplus$)

A blended 4-tuple appears only when Mix *reads* the bowl. Write $\text{Mix} = \boxplus$. Mix is a *phasor* operator: it combines χ among states that already share a direction $\hat{n}$. Direction belongs to T and to the rotations of Axiom 2.8; Mix never moves an arrow onto a new axis. Mix of states with differing $\hat{n}$ is left open (Section 10.1). The output is $(r', \theta', \phi', \chi')$ with $r' \leq 1$, or black. T_w is a one-argument rotation and phase-shift of an occupant, parameterized by an actor w . Mix is a combination of peer states. It is not a member of the group T_w lives in, and it does not compose with T_w the way two ordinary T 's do. The two give different answers on the same pair: for unit red at $\chi = 0^\circ$ and unit green at $\chi = 120^\circ$ sharing a direction, $T_{\text{green}}(\text{red})$ shifts phase additively to $\chi' = 120^\circ$, while red $\boxplus$ green takes the phasor resultant, $\chi' = 60^\circ$. No choice of actor makes T return 60° here, because T adds phase where Mix sums phasors. Three further properties separate them: T_w is asymmetric in actor and patient while $\boxplus$ is commutative; T_w leaves r invariant while $\boxplus$ may grow it to the cap (Axiom 2.5); and every T_w has an inverse w^* while $\boxplus$ has none.

Which sorts may read a Mix result as a claim about their own domain is settled by Axiom 2.1, not here: Tier 1 for **color** and **note-pitch**, inert geometry for **word**, type-refused for **num**.

Two states at a common direction. Treat strength and phase as a phasor $z = re^{i\chi}$. Then

$$z' = r_A e^{i\chi_A} + r_B e^{i\chi_B}, \quad r' = \min(1, |z'|), \quad \chi' = \arg(z'), \quad (9)$$

and $(\theta', \phi') = (\theta_A, \phi_A)$. If $z' = 0$, the result is black. Equal unit paints at 0° and 120° give $z' = e^{i\pi/3}$, so

$$(1\angle 0^\circ) \boxplus (1\angle 120^\circ) = (1, \theta, \phi, 60^\circ) \quad (\text{yellow}).$$

This paper's color inclusion (Axiom 2.1, Tier 1) is that phasor Mix on a toy $0^\circ/120^\circ/60^\circ$ chart. It is not CIE, and it is not linear RGB. The unequal pair $r = 0.6$ at 0° and $r = 0.8$ at 120° gives $z' = 0.20 + 0.6928i$, hence

$$r' \approx 0.721, \quad \chi' \approx 73.9^\circ.$$

Equal anti-phase marks, same place, give $z' = 0$ (black). A state mixed with itself at the same phase is not idempotent: $A \boxplus A$ gives $z' = 2re^{i\chi}$, so $r' = \min(1, 2r)$. Strength grows until the cap clips it. That is distinct from $A \oplus A = A$.

Reading a whole bowl. Mix over a bowl B whose members share a direction $\hat{n}$ is defined in one step, not as a fold:

$$\boxplus(B) = (\min(1, |z_B|), \hat{n}, \arg z_B), \quad z_B = \sum_{x \in B} r_x e^{i\chi_x}. \quad (10)$$

The binary $\boxplus$ of Equation (9) is the case $|B| = 2$. The one-step form is necessary, not stylistic, because the cap makes the binary operator non-associative: with $A = 1\angle 0^\circ$, $B = 1\angle 0^\circ$, $C = 1\angle 180^\circ$,

$$(A \boxplus B) \boxplus C = \text{black}, \quad A \boxplus (B \boxplus C) = 1\angle 0^\circ,$$

since $A \boxplus B$ is clipped from $2\angle 0^\circ$ down to $1\angle 0^\circ$, and the strength discarded by that clip is exactly what C would otherwise have cancelled. Equation (10) on the same three states sums first and caps once:

$$z_B = e^{i0} + e^{i0} + e^{i\pi} = 1, \quad \boxplus(\{A, B, C\}) = (1, \hat{n}, 0^\circ),$$

one reading, reached without choosing an order. $\boxplus$ is commutative, and Equation (10) is order-independent, so a bowl — which carries no order — has exactly one Mix reading. Iterated binary Mix does not, and is not what Equation (10) means. Non-associativity is the price of the cap of Axiom 2.3: without $\min(1, \cdot)$ the phasor sum would be ordinary complex addition and associativity would be free.

$\oplus$ does not run this rule. Two shells in a bowl have no resultant angle until Mix (or a T -path) is applied. A reading returns a state and leaves the bowl intact: $\boxplus(B)$ does not consume B , and a T -path draws actors from a bowl without removing them. Only $\ominus$ takes a member out. That is what makes one bowl readable more than once — the same B may yield a Mix result and, separately, a walked path. Gather is lossless and is undone by $\ominus$; Mix is lossy, since it discards both order and any strength above the cap. That is why $\ominus$ inverts $\oplus$, w^* inverts T_w , and nothing inverts $\boxplus$.

Saturation and levels. A sum carries no bound of its own; the cap of Axiom 2.3 supplies one. A bowl of many agreeing units therefore reaches $r' = 1$ quickly, and further agreement past that point is not recorded. This is worth reading as a remark about levels rather than a defect in the cap. A phasor sum is a reading of units that stand at one level together, and repeated samples of a single unit do not stand at one level; a bowl that saturates early is often a level that has been flattened (Section 2.2). Strength does not accumulate across levels: a reading returns a state whose r is legal by construction — $\min(1, |z_B|)$ for Mix, and the occupant's own r for a path — and that state is seated afresh at the level above. Composition hides the question, since a path over many actors stays in $\text{SO}(3)$ and never saturates, so it is Mix rather than the walk that makes a flattened level visible. What the cap does not supply is a way to average, which needs the count that $\oplus$ discards (Section 2.6); reducing repeated samples to one seated unit is therefore work for the level below, not for $\boxplus$.

2.8 A worked Tier-1 inclusion: chladni

Color is admitted on a declared toy chart. A second sort is admitted here on stronger ground, to show that Equation (1) can be met exactly rather than approximately, and by physics unrelated to either color or language.

Native law. A thin plate driven at one frequency responds as a superposition of its eigenmodes $\varphi_j(x, y)$, each carrying a complex amplitude $A_j = a_j e^{i\chi_j}$; the visible figure is the nodal set of $w = \sum_j A_j \varphi_j$. So $\circ_{\text{chladni}}$ is complex addition of mode amplitudes. Two drives on the same mode add as complex numbers.

Chart. One state is one mode. Let $\hat{n}$ carry the mode index under any injective map $(m, n) \mapsto (\theta, \phi)$, let χ be that mode's phase, and let $r = a/a_{\max}$ be its amplitude normalized to a declared saturation $a_{\max}$.

The map, and where it is exact. Within one mode, $\hat{n}$ is fixed, and

$$\varphi_{\text{chladni}}(A \circ B) = \varphi_{\text{chladni}}(A) \boxplus \varphi_{\text{chladni}}(B) \quad (11)$$

holds identically while $|z| \leq 1$, because complex addition of amplitudes and phasor addition of (r, χ) are the same operation. Nothing is fitted. Note also that the chart's arbitrariness costs nothing here: same-mode combination never moves $\hat{n}$, so Equation (11) places no constraint on the mode-to-direction map. Two consequences are physical and immediate. Equal amplitudes in anti-phase give $z = 0$: the mode is driven to silence and its figure disappears, which is Equation (9) returning black. Two drives at 0° and 120° give $r = 1$ and $\chi' = 60^\circ$ — the same arithmetic as red $\boxplus$ green = yellow, on unrelated physics.

The restriction, which is physical. Across modes the map is not available, and not for want of effort. Mode space is not closed under addition: a sum of two distinct modes is not a mode, so it cannot be seated as a single state under any chart. Cross-mode coexistence is therefore $\oplus$ — a bowl of modes — and the resulting figure is a reading of that bowl which is not $\boxplus$. This is the open question of Section 10.1 met in a concrete domain, and it suggests the question may be closed negatively for this sort rather than left waiting.

Fidelity claimed, and three limits. Exact in phase and amplitude below saturation, within one mode; undefined across modes. Three limits are stated rather than argued away. The cap $r \leq 1$ is a normalization to a declared $a_{\max}$, not a fact of linear plate theory, so Equation (11) clips above saturation, exactly as the color inclusion does. Direction is a label here and nothing more, so no meaning is claimed for T on a **chladni** state: rotating a mode index is not an operation the domain performs. And the figures most often reproduced are degenerate-mode combinations, which fall on the cross-mode side; what is admitted exactly is the simpler case, not the famous one. A rotation $\mathbf{R} \in \text{SO}(3)$ turns the direction $\hat{n}$ and leaves strength, phase, and sort alone:

$$\mathbf{R} \cdot (r, \hat{n}, \chi, \tau) = (r, \mathbf{R}\hat{n}, \chi, \tau), \quad \|\mathbf{R}\hat{n}\| = \|\hat{n}\| = 1. \quad (12)$$

The unit of truth is preserved because r is not scaled and $\hat{n}$ stays on S^2 . This is the geometric engine of T , not a second blend.

3 Transformation Algebra: From Identity to Expression

3.1 Motivation: Two Layers of Meaning

The operators established so far ($\oplus$, $\ominus$) answer the question *who is present?* Combining Sky, Sun, and Wind under $\oplus$ yields one order-independent bowl — a bag of seats, not a blended hue. Paint is Mix ($\boxplus$), a later reading. Analogies such as Sun-Day-Moon are *not* gathers; they are deferred to the offset operator of Section 4. Gather $\oplus$ does not, and structurally cannot, distinguish a poem in which the sun rises through a wind-blown sky from a poem in which wind extinguishes a darkening sun — both share the identical bowl $R = \text{Sky} \oplus \text{Sun} \oplus \text{Wind}$, since $\oplus$ is commutative and idempotent by Axiom 2.6.

The distinction between these two poems is not a difference in *ingredients*; it is a difference in *how one state acts upon another, and in what order*. This is a *relational* operation, and it requires a second operator, T , layered on top of $\oplus$: where $\oplus$ builds nouns, T builds sentences. A poem is not the bowl R itself, but a specific trajectory traced across $\mathcal{S}$ by successive transformations applied to a register occupant.

3.2 Formal Definition of the Transformation Operator

Rather than introduce T as an unrelated new primitive, it is defined using the same rotational machinery already given in Axiom 2.8: every state $w \in \mathcal{S}$ acts on another state $x \in \mathcal{S}$ by rotating x 's direction about w 's own axis, by an angle proportional to w 's strength, while additively shifting x 's phase by w 's phase:

$$T_w(x) = \left(r_x, \text{Rot}_{\hat{n}_w}(\alpha(r_w))\hat{n}_x, (\chi_x + \chi_w) \bmod 2\pi, \tau_x \right) \quad (13)$$

where $\hat{n}_w$ is the unit axis of w , $\text{Rot}_{\hat{n}}(\alpha) \in \text{SO}(3)$ is the rotation of Axiom 2.8 about that axis by angle α , and $\alpha(r_w) = \pi r_w$ is the simplest monotonic choice mapping strength to rotation angle (a full-strength unit, $r_w = 1$, rotates by the maximal 180°). Magnitude r_x and sort τ_x are left invariant: a transformation reorients and recolors a state, it does not strengthen or weaken it, and it does not recast it — those remain the roles of Mix (strength) and of an explicit cast (sort).

3.3 Derived Properties

The following properties follow from Equation (13) rather than being separately postulated.

Identity. If $w = \mathbf{0}$ (the Null State of Axiom 2.4), then $\alpha(0) = 0$ and the phase shift is 0, so $T_{\mathbf{0}} = \text{id}$.

Invertibility. The inverse of T_w is not “the same axis and the same $+\alpha$ again.” Same axis, same positive angle, repeats the turn. Inverse is $-\alpha$ on the same axis, or $+\alpha$ on the antipodal axis, together with $-\chi_w$. Define

$$w^* = (r_w, -\hat{n}_w, -\chi_w \bmod 2\pi, \tau_w), \quad (14)$$

where $-\hat{n}_w$ is the antipode ($\pi - \theta_w, \phi_w + \pi$). Then $T_{w^*}(T_w(x)) = x$ for all x . As rotations, only $\alpha = 0$ ($r_w = 0$) and $\alpha = 180^\circ$ ($r_w = 1$) equal their own reverse; the inverse state w^* is still required so that phase is cancelled too. This undoes a transformation on a state that persists. It is distinct from $\{A\} \ominus A = \emptyset$ (last shell removed, leaving the empty bowl) and from Mix of equal anti-phase marks (a reading into black). Neither of those is an inverse of T .

Associativity. Since each T_w is a function $\mathcal{S} \rightarrow \mathcal{S}$, composition of transformations is associative automatically, by the associativity of function composition; no additional axiom is required.

Non-commutativity. Rotations about distinct axes in $\text{SO}(3)$ generically do not commute: for a, b with distinct directions (θ, ϕ) ,

$$T_a(T_b(x)) \neq T_b(T_a(x)). \quad (15)$$

This is not an additional assumption; it is inherited directly from the well-established non-commutativity of $\text{SO}(3)$, already invoked by Axiom 2.8. It is this property, and only this property, that allows two poems built from the same bowl to diverge: order of application is meaningful even though the underlying identity is not.

3.4 Poems as Transformation Trajectories

A poem π built from a register occupant x and an ordered sequence of transforming states $w_1, w_2, \dots, w_n$ (typically drawn from the bowl) is the trajectory

$$\pi = (x; w_1, \dots, w_n) : \quad x \longrightarrow T_{w_1}(x) \longrightarrow T_{w_2}T_{w_1}(x) \longrightarrow \dots \longrightarrow T_{w_n} \dots T_{w_1}(x). \quad (16)$$

Two poems may share an identical bowl (the same theme, the same “ingredients”) while tracing entirely different trajectories — this is the formal sense in which many poems arise from the same units of truth. Conversely, because rotation composition is many-to-one, two distinct trajectories may terminate at the same endpoint; a poem’s identity is therefore carried by its full path through $\mathcal{S}$, not by its final state alone.

3.5 Worked Numerical Example: Sun, Wind, and Sky

Let Sky be the state acted upon, with direction $(\theta, \phi) = (45^\circ, 45^\circ)$, i.e. Cartesian axis $(0.500, 0.500, 0.707)$. Let Sun define an axis along $(\theta, \phi) = (90^\circ, 0^\circ)$, i.e. $(1, 0, 0)$, with strength $r_{\text{Sun}} = 1.0$, giving rotation angle $\alpha_{\text{Sun}} = 180^\circ$. Let Wind define an axis along $(\theta, \phi) = (90^\circ, 90^\circ)$, i.e. $(0, 1, 0)$, with strength $r_{\text{Wind}} = 0.7$, giving $\alpha_{\text{Wind}} = 126^\circ$.

Order 1 — Sun acts first, then Wind (the sun rising through a wind-blown sky):

$$T_{\text{Wind}}(T_{\text{Sun}}(\text{Sky})) \approx (-0.866, -0.500, 0.011) \quad (17)$$

Order 2 — Wind acts first, then Sun (wind extinguishing a darkening sun):

$$T_{\text{Sun}}(T_{\text{Wind}}(\text{Sky})) \approx (0.278, -0.500, 0.820) \quad (18)$$

The two endpoints are clearly distinct, confirming Equation (15): identical primitives, identical bowl, opposite structural readings. Notably, if $\chi_{\text{Sun}} = 0^\circ$ and $\chi_{\text{Wind}} = 90^\circ$, the accumulated phase shift is 270° in *both* orderings, since phase addition is commutative. The two poems therefore share the same blended color/mood even as their spatial — structural — meaning diverges. This asymmetry (shared affect, divergent structure) is a numerical check of the algebra, not a theory of verse.

3.6 Scope

T is deliberately left unconstrained beyond Equation (13): any sequence of any transforming states is a formally valid trajectory. Pattern Arithmetic in this form supplies the substrate on which expression is built — the combinatorial space of possible trajectories — and does not yet supply a grammar constraining which trajectories are coherent, well-formed, or meaningful to a human reader. Whether such constraints belong inside this framework (as further axioms on admissible sequences of T) or outside it (as a learned or externally specified grammar layered on top) is left open.

4 Analogical Arithmetic: The Offset Operator

Offset is another verb. It is not $\oplus$ (no gather), not Mix (no phasor blend), and not T (no spin of an actor on a path). It copies a *walk* from one seat to another. The landing is a new direction on the same globe. A dictionary may later pin a name on that landing. The walk does not mint the name.

4.1 Recipe: logarithm, carry, exponential

Write directions as unit vectors $\hat{n} \in S^2 \subset \mathbb{R}^3$. The walk from p to q is the tangent vector

$$\theta = \arccos(\hat{p} \cdot \hat{q}), \quad \log_{\hat{p}}(\hat{q}) = \frac{\theta}{\sin \theta} (\hat{q} - (\hat{p} \cdot \hat{q}) \hat{p}) \quad (\theta \neq 0), \quad (19)$$

with $\log_{\hat{p}}(\hat{p}) = \mathbf{0}$. Its length is the geodesic angle. The reverse map, from a tangent vector v at p , is

$$\exp_{\hat{p}}(v) = \cos \|v\| \hat{p} + \sin \|v\| \frac{v}{\|v\|} \quad (v \neq \mathbf{0}), \quad (20)$$

and $\exp_{\hat{p}}(\mathbf{0}) = \hat{p}$. Given three seats m, k, w (source, source-image, new start), set

$$v = \log_{\hat{n}_m}(\hat{n}_k), \quad \hat{n}_* = \exp_{\hat{n}_w}(\text{transport}_{m \rightarrow w}(v)). \quad (21)$$

The carry rotates the tangent vector by the geodesic angle about the axis of $m \rightarrow w$:

$$\theta_{mw} = \arccos(\hat{n}_m \cdot \hat{n}_w), \quad \hat{u} = \frac{\hat{n}_m \times \hat{n}_w}{\|\hat{n}_m \times \hat{n}_w\|} \quad (\hat{n}_m \neq \pm \hat{n}_w), \quad (22)$$

$$\text{transport}_{m \rightarrow w}(v) = \text{Rot}_{\hat{u}}(\theta_{mw}) v = v \cos \theta_{mw} + (\hat{u} \times v) \sin \theta_{mw} + \hat{u} (\hat{u} \cdot v) (1 - \cos \theta_{mw}). \quad (23)$$

The axis is undefined when $\hat{n}_m = \pm \hat{n}_w$. Coincident seats ($\hat{n}_m = \hat{n}_w$): the carry is v itself. Antipodal seats ($\hat{n}_m = -\hat{n}_w$): no unique short geodesic; that case is a separate convention and is not used below. For a first recipe, if m and w are not far, transport may be omitted and v used at w as written. That case is crude and marked as such. The full recipe is (23), so the heading stays a ground heading.

Strength and colour are not invented by the walk. This paper takes

$$r_* = r_w, \quad \chi_* = (\chi_w + \chi_k - \chi_m) \bmod 2\pi, \quad \tau_* = \tau_w. \quad (24)$$

The target keeps its r and its sort. The mood increment of the pair $m \rightarrow k$ is reused. Nothing here uses $\alpha = \pi r$ from T . With (21) and (24) together, offset returns a complete state, not a bare direction:

$$\text{offset} : \mathcal{S} \times \mathcal{S} \times \mathcal{S} \longrightarrow \mathcal{S}, \quad (m, k, w) \mapsto (r_w, \hat{n}_*, \chi_*, \tau_w).$$

It is therefore a verb on the same footing as $\oplus$, $\boxplus$ and T : a landing may be gathered into a bowl, walked from, or used as the start seat of a further offset. The question of *whether anything is seated there* is separate, and is settled by a dictionary, not by the walk. $\tau_* = \tau_w$ because the question asked is what stands to w as k stands to m ; the answer is of w 's kind. A walk copied between sorts would need an explicit cast (Section 5), which offset does not perform. Two seated units keep two strengths. If Day has $r = 0.9$ and Night has $r = 0.8$, both sit; there is no $0.9 + 0.8 = 1.7$. That sum is ambient $\mathbb{R}^3$ arithmetic and leaves the ball. $\oplus$ gathers the two seats. A later path may turn a dial or clip a mix. It does not get to smash two r 's into one illegal length as the price of seating them.

4.2 A check on one equator, not a definition

Place three arrows in the xy -plane:

$$\hat{n}_m = (1, 0, 0), \quad \hat{n}_k = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}, 0\right), \quad \hat{n}_w = (0, 1, 0)$$

$(0^\circ, 30^\circ, 90^\circ)$. Then $\log_{\hat{n}_m}(\hat{n}_k) = (0, \pi/6, 0)$. Here $\theta_{mw} = 90^\circ$ and $\hat{u} = (0, 0, 1)$, so (23) is a quarter-turn about $\hat{z}$ and gives $(-\pi/6, 0, 0)$. The exponential at $\hat{n}_w$ lands at

$$\hat{n}_* = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right) \quad (120^\circ).$$

On this slice the walk *looks like* $90^\circ + 30^\circ = 120^\circ$. That sum is a protractor check. It is not the operator, and it is not a night. If a lexicon writes the names Sun, Day, Moon on the first three seats, it may write Night on the landing: whatever Sun is to Day, Moon is to that fourth place. The geometry only promised a fourth pin. The same moral as weather: Day and Night are dictionary pins, not sums of luminaries. The step that carries Sun to Day is a walk, and the walk is what is copied; $\oplus$ of those seats does not land on a time of day. Whether the two halves really are the same walk depends entirely on how a lexicon seated them, which is a dictionary claim and not one this paper makes. Offset only copies a walk; a later comparison may find the landing *near* an already seated name. It does not mint the name.

4.3 Offset is not T

$T_w(x)$ spins an occupant about an axis. Offset plants a new seat from three seats. A path does not need (21). A dictionary act does not need (13). Do not encode the walk as a ghost actor whose sole job is to trigger T .

The popular ambient sum $\hat{n}_k - \hat{n}_m + \hat{n}_w$ then normalize (Recipe 1) is the embedding fluke written as geometry. It is recorded as a cheaper cousin, not as a derivation of the landing. This paper uses Recipe 2 so the copied step stays on the skin of the ball.

5 Encodings, Sorts, and Mixed Documents

The unit sphere is a single bounded *register*. Numbers, words, and colors are not three different globes; they are three encodings of a token onto the same globe. A scientific paragraph already mixes those alphabets on one page. Pattern Arithmetic may do the same, provided each loaded state carries a *sort* and the operators respect that sort.

5.1 Two primary sorts

A state $x \in \mathcal{S}$ is the typed atom of Axiom 2.3. What the geometric 4-tuple (r, θ, ϕ, χ) *means* depends on the sort τ and the encoding used to write it:

- **Numeric sort (num).** The state represents a scalar. One toy encoding — the only numeric encoding used in this paper — stores an integer n on a marked clock as $\chi = 2\pi n/N$ (arithmetic modulo N), at a fixed direction, with $r = 1$. Another, stereo-style, is recorded as an option: it would send $x \rightarrow \pm\infty$ toward the poles so that unbounded reals still occupy the globe. It is not used below.

- **Lexical sort (word).** The state is a labeled point in a toy dictionary living on the same globe. Similarity is the angle between points (cosine similarity). A bag of words is formed by $\oplus$. A sentence or poem is formed by T .

Schoolbook addition lives only in **num** mode. Mix as a *domain* reading lives in the Tier-1 sorts of Axiom 2.1: **color** under this paper’s phasor inclusion, and **note-pitch**, whose octave equivalence makes the circle native rather than borrowed. The two sorts worked below are **num** and **word**; neither takes a domain Mix. Two **word** states may be mixed as geometry; that is not a χ -reading of language. The two equations may share a sphere. They may not share an operator.

5.2 Numeric mode: $7 + 5 = 12$ on the clock, and why $\oplus$ is not 12

Fix $N = 24$ and the encoding

$$E_{\text{num}}(k) = (r = 1, \hat{n} = (1, 0, 0), \chi = 2\pi k/24, \tau = \text{num}), \quad k \in \mathbb{Z}/24\mathbb{Z}. \quad (25)$$

Decode is the inverse chart: $k = N\chi/(2\pi)$, an integer hour mark. Then 7 sits at $\chi = 105^\circ$, 5 at $\chi = 75^\circ$, and 12 at $\chi = 180^\circ$. Schoolbook $+$ is decoded arithmetic:

$$7 + 5 = E_{\text{num}}^{-1}(E_{\text{num}}(7)) + E_{\text{num}}^{-1}(E_{\text{num}}(5)) = 12, \quad E_{\text{num}}(12) = (1, (1, 0, 0), 180^\circ, \text{num}). \quad (26)$$

That is $7 + 5 = 12$ as a calculation on $\mathcal{S}$: decode, add in $\mathbb{Z}$, encode. Gather is legal on the same pair but answers a different question: $7 \oplus 5 = \{7, 5\}$ records which numerals are present and is not 12. If one instead applies Mix (Equation (9)) to the same two geometric marks, the phasors are

$$z = e^{i7\pi/12} + e^{i5\pi/12} = 2\cos(\pi/12)e^{i\pi/2}, \quad (27)$$

so $r' = \min(1, 2\cos 15^\circ) = 1$ and $\chi' = 90^\circ$. The clock would read that mark as hour 6, not hour 12. A bowl $\{E_{\text{num}}(7), E_{\text{num}}(5)\}$ is two seats, still not 12. This single contrast is why the type table exists: $+$ answers *how many*; $\oplus$ and Mix do not.

5.3 A document is a sequence, not a blend

Words and numbers coexist in reports because a document is an ordered tape of tokens, not one mixed arrow. Write a paragraph as a sequence of typed atoms

$$(x_1, \dots, x_m), \quad x_i = (r_i, \hat{n}_i, \chi_i, \tau_i), \quad \tau_i \in \{\text{num}, \text{word}\}, \quad (28)$$

The sphere holds one current occupant of the register; the paragraph is the history of loads and actions. A poem (Section 3.4) is one species of that history: a T -walk on one occupant. Juxtaposition is another. Mix is not.

They may meet in only three honest ways.

Juxtaposition. The default. Tokens sit in order and no algebra is applied between them. In “the sample reached 37.4°C at dawn,” the number 37.4 is not blended with the word dawn.

Typed action. The operator $T_w(x)$ is allowed across sorts only with an explicit output sort. Two useful conventions, which the rest of this paper adopts, are:

- a word acting on a number yields a number-in-context (still **num**);
- a number acting on a word yields a refined word (still **word**).

A number may also set a numeric dial of T — for example the rotation angle $\alpha(r)$ — because that dial is already a scalar. That is how “wind of 12 m/s” can strengthen a turn without becoming a synonym of wind.

Cast. Rare and named. The map $E_{\text{num} \rightarrow \text{word}}(2) = \text{two}$ turns a digit into a spoken numeral; the inverse is parsing. Casts are dictionary maps, not $\oplus$.

The illegal meeting is an untyped blend

$$\text{dawn} \oplus 37.4, \tag{29}$$

which has no meaning until one side is first cast into the other’s sort.

5.4 Operator legality

Table 2 records which buttons accept which pairs. Here $+$ stands for schoolbook arithmetic on the numeric encoding (decode, operate in $\mathbb{Z}$, encode; $+$ is the worked case of Section 5.2), $\oplus$ is idempotent gather, $\boxplus$ is Mix, T is typed action, and $x - a + b$ is the analogical offset of Section 4.

5.5 Worked mixed sentence

Take the report line *Wind of 12 m/s darkened the sun.* The register is used as follows.

1. Load wind, sort **word**.
2. Load 12, sort **num**. Juxtapose; do not form $\text{wind} \oplus 12$.
3. Load m/s as a unit-name (lexical qualifier of the number).
4. Load sun, sort **word**.
5. Let the number cap a strength, $\sigma(12) \in [0, 1]$, and set $r_{\text{wind}} = \sigma(12)$ so that $\alpha = \pi\sigma(12)$ in Equation (13).
6. Apply $T_{\text{wind}}(\text{sun})$. Output sort: **word**.

The scalar never became a hue. It turned a dial of an action whose actors remained words. That is the same division of labor a printed paper already uses: digits set magnitudes; words name what the magnitudes belong to.

Pair	+	$\oplus$	$\boxplus$	$T_w(x)$	$x - a + b$
(word, word)	no	yes	yes [†]	yes	yes
(num, num)	yes	yes [‡]	no	no*	no*
(word, num)	no	yes [‡]	no	typed	no
(num, word)	no	yes [‡]	no	typed	no
(color, color)	no	yes	yes	yes	open
(note-pitch, note-pitch)	no	yes	yes	yes	open
(chladni, chladni)	no	yes	yes [§]	open	open
(τ_1, τ_2) any other	no	yes [‡]	open	open	open

Table 2: Sort discipline on a single sphere. Every $\boxplus$ entry is additionally subject to the place restriction of Axiom 2.7: Mix combines χ at a common $\hat{n}$, and Mix across differing directions is open (Section 10.1). **color** and **note-pitch** are the Tier-1 sorts of Table 1; they gather like any other sort, and their Mix is a domain reading rather than inert geometry. Mix on (num, num) is the clock trap of Section 5.2: it is geometrically defined and type-refused. [†]Mix on (word, word) is allowed as geometry and carries no linguistic χ -claim (Axiom 2.1, Tier 0). *Numeric T and numeric offset may be defined later as maps of the chosen numeric encoding; they are not the lexical operators of Sections 3 and 4. Offset on the Tier-1 sorts is untested here and marked open rather than claimed. [‡]Gather is sort-blind: the $\oplus$ column is *yes* for every pair, including pairs of different sorts and sorts not listed here. A bowl records which units are present, and presence carries no domain claim, so there is nothing for a sort mismatch to corrupt. What gather does *not* do is combine: $2 \oplus \text{two} = \{2, \text{two}\}$ is two seats and no cast, and $1 \oplus 1 = \{1\}$ is one seat and not the sum 2. Counting stays with $+$ (Section 5.2). The final row is the general rule; a sort seated on the sphere may always be gathered, while its other operators wait on Axiom 2.1. Sorts with no chart at all, such as **snowflake** in Table 1, have no row here: nothing is seated, so nothing is gathered. [§]Mix on (chladni, chladni) is a domain reading only within one mode, where the place restriction above is satisfied automatically; across modes it is undefined rather than merely untested (Section 2.8). T is marked open because direction carries a mode label for this sort and no domain meaning is claimed for rotating it.

5.6 Several lexicons, one concept sphere

Surface tokens are not meanings. Two dictionaries may write the same pair of concepts with unrelated strings:

$$\begin{aligned}
L_1 : \quad & \text{xyz} \mapsto \text{man}, \quad \text{qyz} \mapsto \text{woman}, \\
L_2 : \quad & \text{fgh} \mapsto \text{man}, \quad \text{hgf} \mapsto \text{woman}.
\end{aligned} \tag{30}$$

In the usual pipeline each lexicon is tokenized, encoded, and *embedded separately*. The geometry that makes man near woman is then learned twice, and a further alignment step is required before L_1 and L_2 can share an analogy. The token-to-embedding map is doing two jobs at once: naming a string, and inventing a place for its meaning.

The unit sphere is offered as a *proposed common coordinate register*, not as a demonstrated universal semantic space and not as a replacement for a lexicon table. A token is a surface form. A state on $\mathcal{S}$ is a meaning. Each lexicon is only a cast

$$E_L(\text{token}) = x \in \mathcal{S}, \tag{31}$$

the same kind of map as $E_{\text{num} \rightarrow \text{word}}$ above. Under that discipline both **xyz** and **fgh** load the *same* state **man**. The offset **Sun** $\rightarrow$ **Day** of Section 4 is then computed once, on the globe, and reused by every lexicon that points at those two points.

Embeddings therefore stay. They do not disappear. What changes is that they are no longer private vector spaces, one per lexicon. They are embeddings *into the same unit* $\mathcal{S}$. Once $E_{L_1}(\text{xyz}) = E_{L_2}(\text{fgh}) = \text{man}$, the two surface forms occupy one address, and every angle measured from that address is the same angle:

$$\angle(\text{man}, \text{woman}) = \angle(E_{L_1}(\text{xyz}), E_{L_1}(\text{qyz})) = \angle(E_{L_2}(\text{fgh}), E_{L_2}(\text{hgf})). \quad (32)$$

Under that proposal the embeddings would be uniform: same globe, same points for the same concepts, same angles for the same relations. A raw string still needs the lexicon map E_L . It would no longer need a new geometry. The algebra does not mint a universal **man** point by itself. That is a dictionary programme, not an axiom.

5.7 Deferred programme: standardized concept dictionaries

The present section only *names* the shared globe. It does not build the dictionaries, and it does not treat them as part of the algebra closed in Axioms 0–6. The later refinement, if the idea is pursued, is a pair of catalogues rather than a new operator, and is filed under Future Work (Section 10):

- one *concept dictionary* — a stipulated inventory of unit states on $\mathcal{S}$ (or a higher-dimensional analogue), so that **man** and **woman** have standard addresses and therefore standard angles;
- one *language lexicon* E_L per written or spoken system, sending that language’s tokens onto those addresses;
- learnable paths — the species of T and the dials inside a given T (angle map, phase map, paint-versus-poem gate) may be fitted from examples. Learning would adjust the path, not the rule that a bowl is collected before it is read. See also Section 10.

Standardization of the first catalogue is what makes embeddings uniform across English, a cipher, or a second natural language. The second catalogue is still a tokenizer-plus-encoding: it is the phone book, not the city. Tokens and encodings have already earned their keep, so they stay. They only have to work *lighter*: look up a standard address instead of inventing one. They disappear as *inventors of geometry*. They do not disappear as *names*. A world in which every language pointed at the same concept dictionary would still print letters on a page; It would no longer need each corpus to learn a private **man**–**woman** angle. That programme is recorded here and left open. It is not an axiom, and it is not implied by the existence of $\mathcal{S}$ alone.

5.8 Scope of the register model

This section claims only that one sphere can *host* both alphabets as a workspace, and that several lexicons can point at the same workspace through different casts. It does not claim that the two-dimensional surface S^2 is a realistic embedding space for a full natural-language lexicon: a working word embedding typically needs tens or hundreds of dimensions. The construction here is a typed register and a toy dictionary on one

globe. Scaling the lexical encoding is a later choice — a higher-dimensional sphere, a spherical-harmonic field, or an external lookup table that writes states into the register. The lookup tables E_L may still be learned; the claim is only that they should write into one geometry rather than each minting a private one. The unit ball of radius 1 in $\mathbb{R}^3$ is where the idea is taught. The unit sphere in the embedding dimension is where close semantics can live. The algebra — collapse, path, result — is meant to travel with the dimension. It is not meant to replace 768 numbers with two angles and hope the library still sorts.

6 A Sketch of Conventions

These notes are a seed for later grammars, not a code to enforce. They show one way a typed register *might* be spoken. A later grammar — learned, cultural, or formal — may replace every line. This paper only needs the ground level enough that such a grammar has somewhere to stand.

Case does not move the address. The tokens **man**, **Man**, and **MAN** load the same direction (θ, ϕ) . Orthography is a doorbell, not a new city block.

Language does not move the address. English **man** and a Polish token for the same concept load the same point. Only the lexicon map E_L changes. Language is a phone-book tag L , not a second angle.

Color is reserved for mood, not for language. Phase χ is already used as affect in the poem example (Section 3.5). If language or case also occupied χ , that example would break. Emphasis may tint χ or raise r ; it should not relocate Sun to Moon.

A bag is not a sentence. $\{\text{sun}, \text{wind}\}$ is $\oplus$. “Wind darkens the sun” is T . Ingredients and order are different verbs. How often a unit is used is not a second shell in the bag: the path may visit **twinkle** twice; **num** may count $1 + 1 + 1 + 1 = 4$. $\oplus$ will not.

A numeral is not a word until it is cast. The digit 2 stays **num**. The word **two** is **word**. They may share a bowl — gather is sort-blind — but $2 \oplus \text{two} = \{2, \text{two}\}$ is two seats, not a cast and not a sum. Becoming a word requires the explicit $E_{\text{num} \rightarrow \text{word}}$ of Section 5, which no gather performs.

A toy row makes the first three conventions visible. Place **man** at $(\theta, \phi) = (10^\circ, 40^\circ)$:

Token	L	(θ, ϕ)	χ (mood only)
man	English	$(10^\circ, 40^\circ)$	0° (neutral)
Man	English	$(10^\circ, 40^\circ)$	0°
MAN	English	$(10^\circ, 40^\circ)$	20° (emphasis, optional)
man	Polish	$(10^\circ, 40^\circ)$	0°

Same block. Different doorbells. Nothing here is binding.

7 Operational Algebra & Numerical Demonstration

7.1 Mix of two shells

Collecting $r = 0.6$ at 0° with $r = 0.8$ at 120° (same place) yields two shells, not an angle. Applying Mix ($\boxplus$) and Equation (9) yields the 4-tuple $(r', \theta', \phi', \chi') \approx (0.721, \theta, \phi, 73.9^\circ)$. The older figure 66.6° is withdrawn: Axiom 5 as a length-only formula never produced it.

7.2 Rotational transposition, and what division is not

Rotation is Axiom 2.8: $\mathbf{R} \in \text{SO}(3)$ turns $\hat{n}$. Inverse rotation is well-defined. Division as “deconvolution and signal unmixing” is not defined in this paper and is left open.

7.3 Spectral storage (optional, later)

Spherical-harmonic arrays of size $O(L^2)$ may later store a field of many atoms. They are not the working atom of Axiom 2.3, and $O(N^3) \rightarrow O(L^2)$ is not a theorem of Pattern Arithmetic as it stands.

8 Higher-order systems as bowls and paths

These examples illustrate the $\oplus$ -versus- T split with borrowed vocabulary. They are costumes (Section 1.4), not models of weather, ecology, or disease. Higher-order truths are bowls of basic units read by a path. A storm, a forest, or a cancellation is such a reading. It is an example of the spine, not a model of the sky, the woods, or the clinic. The path defines the higher-order system: the bowl only lists who is present.

A vertical pinball table makes the same split visible. Each bumper is one seated unit — a direction and a size r . The set of bumpers on the board is the bowl (wind, temperature, humidity, pressure). The pockets at the bottom are dictionary pins already named storm and calm. Dropping a ball from a slot is a path T_{weather} : the ball meets some bumpers, in some order, and each kick depends on that bumper’s angle and strength. The landing pocket is a comparison after the walk. Adding the bumpers together does not *make* a pocket. A different slot, or a different order of kicks, is a different weather. This is one rough picture of *a* transform path, not the mechanics of every T , and not a second axiom. A Mix reading ($\boxplus$), and a poem path T_{poem} , need not bounce.

8.1 Atmospheric storm (costume): one pipeline with numbers

This is a costume for the spine, not a model of the sky. These walks are not weather equations, and they are not forest equations. They are still computations: ordered readings of many co-present units that can land near a higher-order name already seated in a dictionary. Four incoming states, one bowl, two paths, two endpoints. Every figure below is Rodrigues’ rotation of $\hat{n}$ by $\alpha = \pi r$ about the actor’s axis, with χ added. A reader can recompute.

1. Input states.

$$\begin{aligned}
\text{Wind} : r = 1, \hat{n} &= (1, 0, 0), \chi = 0^\circ, \tau = \text{word}, \alpha = 180^\circ, \\
\text{Heat} : r = \frac{1}{2}, \hat{n} &= (0, 1, 0), \chi = 90^\circ, \tau = \text{word}, \alpha = 90^\circ, \\
\text{Moisture} : r = \frac{2}{3}, \hat{n} &= (0, 0, 1), \chi = 45^\circ, \tau = \text{word}, \alpha = 120^\circ, \\
\text{Eye} : r = 1, \hat{n} &= \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right), \chi = 0^\circ, \tau = \text{word}.
\end{aligned}$$

The eye sits at $(\theta, \phi) = (45^\circ, 0^\circ)$.

2. Collapse into the bowl.

$$R = \text{Wind} \oplus \text{Heat} \oplus \text{Moisture} \oplus \text{Eye} = \{\text{Wind}, \text{Heat}, \text{Moisture}, \text{Eye}\}. \quad (33)$$

Four distinct names, four seats. A second Wind would not open a fifth. Call this bowl $s(\text{weather})$ if a name helps: it is still four pins, not a fifth vector. Mix is a different reading and is not the storm. In the dictionary, storm and calm are *other* seated units — separate doorbells, not sums of wind, temperature, humidity, and pressure. Mere $\oplus$ cannot reach those pins. T_{weather} is the path family that reads $s(\text{weather})$; T_{deepen} and T_{clear} below are two such walks. An endpoint may fall *near* the storm pin or the calm pin. Nearness is comparison after the walk, not addition of the parts.

3. Two trajectory paths. The register occupant is the eye. The other three shells act, in order:

$$T_{\text{deepen}} = T_{\text{Wind}} \circ T_{\text{Heat}} \circ T_{\text{Moisture}}, \quad T_{\text{clear}} = T_{\text{Moisture}} \circ T_{\text{Heat}} \circ T_{\text{Wind}}. \quad (34)$$

Deepen — moisture, then heat, then wind:

$$\begin{aligned}
T_{\text{Moisture}}(\text{Eye}) &= (1, (-\frac{\sqrt{2}}{4}, \frac{\sqrt{6}}{4}, \frac{\sqrt{2}}{2}), 45^\circ), \\
T_{\text{Heat}}(\cdots) &= (1, (\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{4}, \frac{\sqrt{2}}{4}), 135^\circ), \\
T_{\text{Wind}}(\cdots) &= (1, (\frac{\sqrt{2}}{2}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{4}), 135^\circ).
\end{aligned}$$

Clear — wind, then heat, then moisture:

$$\begin{aligned}
T_{\text{Wind}}(\text{Eye}) &= (1, (\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}), 0^\circ), \\
T_{\text{Heat}}(\cdots) &= (1, (-\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}), 90^\circ), \\
T_{\text{Moisture}}(\cdots) &= (1, (\frac{\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{2}), 135^\circ).
\end{aligned}$$

4. Endpoints. T leaves the occupant's strength at $r = 1$. Phase addition commutes, so both paths end at $\chi = 135^\circ$. The directions do not agree:

$$\begin{aligned}
T_{\text{deepen}}(\text{Eye}) : \hat{n} &= \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{4}\right) \approx (0.7071, -0.6124, -0.3536), \\
(\theta, \phi) &\approx (110.7^\circ, 319.1^\circ); \\
T_{\text{clear}}(\text{Eye}) : \hat{n} &= \left(\frac{\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{2}\right) \approx (0.3536, -0.6124, -0.7071), \\
(\theta, \phi) &= (135^\circ, 300^\circ).
\end{aligned}$$

Same bowl, two weathers. The storm is the walk. It is not a fifth pin, and it is not a claim about vorticity or shear.

8.2 Table and chair (costume)

A table is one seated unit; a chair is another. $\{table, chair\}$ is their bowl under $\oplus$ — two names present, order-blind, no claim about arrangement. A room is not read from that bowl. It is read from a path: $T_{chair}(table)$, the chair acting on the table, is a placement — pulled up to the table’s edge. $T_{table}(chair)$, the same two shells with the actor swapped, is a different placement — the table built around a fixed chair. $\oplus$ cannot tell these apart; it only knows both are present. T can, because T is not symmetric in its two arguments the way $\oplus$ is. As with the storm’s pockets, a named room — “dining nook,” “reading corner,” “cluttered storage” — is a separate dictionary pin, not a sum of the furniture. An endpoint may land near one pin or another; nearness is comparison after the walk, exactly as in Section 8.1. This is a costume for the same split table and chair make familiar without any physics: two objects merely being present is one reality (grocery counting: one bag of grain plus one bag of grain is two bags — schoolbook $+$, not $\oplus$); two objects placed in a specific relation is a different reality, and no amount of $\oplus$ on the parts reaches it. It is not a model of furniture, interior design, or how rooms are actually named.

The furniture case is also where the motivating hypothesis behind this whole paper is easiest to say plainly: a path T_{chair} that keeps “chair acts on table” as one typed, invertible step is a shorter thing to learn than the same fact buried inside a flat embedding, where the model must first discover, from undifferentiated vectors, that arrangement is a different kind of fact than mere co-occurrence. Keeping presence ($\oplus$) and path (T) as separate primitives is a bet that higher-order truths become easier to learn because the distinction is built in rather than re-derived from data. That bet is stated here as a hypothesis, not a result: no example in this paper demonstrates it, and one is deliberately not attempted. Section 10.1 leaves it open.

8.3 Forest (costume)

Flora, fauna, soil, moisture sit as units. $\oplus$ gathers them into one bowl — four names present, order-blind (Axiom 2.6). There is no $A+B=AB$. A path may read competition or succession. The forest-as-system is that reading.

8.4 Pathogen cancellation (costume)

A mark and an equal-strength anti-phase mark gather as two seats in one bowl (Axiom 2.6); $\oplus$ does not cancel them. That bowl may then be *read* by Mix into black (Axiom 2.7). That is geometry. It is not medicine.

9 Discussion & Related Work

Each paragraph below states what is reused and what is added. None of these citations is a claim of identity. The list is a boundary-drawing exercise against formalisms that share a sphere, an angle, or a rotation (Section 1.4). It is not a unification of those fields, and it is not a proposal to replace quantum mechanics, embeddings, or transformers. Empty headings are omitted.

Bloch sphere. A pure qubit is a point on S^2 ; a mixed qubit lives in the ball. Pattern Arithmetic reuses that unit globe. It differs in three respects. Strength $r \leq 1$ is an activation of one arrow, not a purity parameter of a density matrix. Phase χ is an internal colour of that arrow, not a global $U(1)$ that can be quotiented away. There is no measurement postulate: Mix may mint a third unit or black; $\oplus$ only collects. They do not collapse a superposition onto an eigenstate. Related work may say: we reuse a unit globe; we do not borrow quantum authority for $\oplus$.

Word2vec and cosine embeddings. Lexical similarity as an angle is reused. A bag of words as a commutative gather is reused in spirit. What is added is a bounded unit ($r \leq 1$), an explicit collection operator that is idempotent ($A \oplus A = A$), and an order-dependent path T that cosine addition does not supply. Offset is a manifold cousin of the familiar king – man + woman arithmetic, not a proof that the toy globe is a full lexicon.

Transformers. A transformer occupies the same *slot* as T : it is a path through seated tokens. It is not the same device. The usual stack is

sentence $\rightarrow$ token $\rightarrow$ encode $\rightarrow$ embed $\rightarrow$ transformer.

The stack of this paper is

sentence $\rightarrow$ token $\rightarrow E_L \rightarrow$ sphere $\rightarrow \oplus$ (bowl) + T (walk).

Tokens and encodings stay. The embedding is a cast into one sphere (of whatever dimension the lexicon needs). T_{poem} , T_{prose} , and T_{equation} are species of walk, not extra nets. A transformer may later fill those walks as a learned path. This paper does not claim to replace attention; it splits presence from order so that a learned path need not also invent the city.

RotatE. Knowledge-graph embeddings that treat a relation as a rotation in a complex plane are the nearest neighbour of T . Pattern Arithmetic extends that idea from S^1 (or a product of circles) to $SO(3)$ acting on S^2 , and pairs the rotation with an additive phase χ . It also keeps a separate collapse operator, which RotatE does not.

von Mises–Fisher. Directional statistics on the sphere reuse the same stage. Those models estimate a density. Pattern Arithmetic does not: the atom is a point, r is not a concentration κ , and $\oplus$, $\boxplus$, T , and offset are algebraic operations on units, not maximum-likelihood fits.

Riemann sphere. Stereographic compactification of $\mathbb{C} \cup \{\infty\}$ onto S^2 is the natural home of one numeric encoding (unbounded reals toward the poles). The algebra of this paper is not the Möbius group. A clock, or a stereo map, is a chart; $+$ in **num** mode is decoded arithmetic, not a fractional linear map.

$SO(3)$. The transformation T is a named action of the rotation group, plus a phase shift. That is the whole geometric engine of a path. What $SO(3)$ does not supply is the bowl, the cap $r \leq 1$, the sort τ , or Mix. Non-commutativity of poems is inherited; idempotence of $\oplus$ is not.

Spherical harmonics. A later field representation of many arrows could be stored as Y_ℓ^m coefficients, replacing an $O(N^3)$ voxel cube by an $O(L^2)$ spectral array. That is a compression option for a field, not a definition of the atom, and it is not claimed as a result here.

10 Conclusion & Future Directions

The claim of this paper is narrow (Section 1.4). Counting is not the only arithmetic for combining units of information. A combination rule should match the unit. The sphere and the operators $\oplus$, $\boxplus$, T , and offset are one proposed encoding of that claim. They are not a general theory of language, physics, biology, or computation. The costumes and related-work paragraphs are illustrations and boundaries, not domain models.

What was written is a single pipeline: collapse many sorted units into a bowl of typed atoms $x = (r, \hat{n}, \chi, \tau)$ on a unit sphere, then derive results by ordered transformation of a register occupant. Embeddings, on this view, are not abolished. They are embeddings into one proposed common coordinate register, so that a concept such as man *may* keep the same angles under every lexicon that points at it. That is a design, not a demonstrated universal semantic space.

A few spoken conventions are sketched in Section 6 as seed, not statute.

10.1 Future work

What remains to be built, and is only filed here, is a standardized concept dictionary together with per-language lexica E_L (Section 5.7). Until those catalogues exist, tokenization and encoding remain the handles by which a paragraph finds its points. They may continue; they will only have to work lighter. The open bet is that they need not also remain the architects of a new space of meanings. The algebra closed in this paper does not produce a universal man point by itself. Standardization is sociological and empirical; it is not an axiom.

A further deferred question is whether a path T should stay hand-specified or carry learnable weights: the angle law $\alpha(r)$, a phase map, a gate between Mix ($\boxplus$) and T_{poem} , and the lexicon casts E_L . The honest constraint is structural. If undo-the-poem matters, learning should stay inside $\text{SO}(3)$, not collapse T into an arbitrary matrix. Weights may change how the shells are thrown. They should not let the shells throw themselves.

Left fully open is the hypothesis raised in Section 8.2: that separating presence ($\oplus$) from path (T) as distinct primitives should make higher-order truths cheaper to learn than recovering the same distinction from undifferentiated vectors. Testing this would need a controlled comparison — a compositional task, a flat-embedding baseline, and a T -path model, measured on data efficiency rather than final accuracy — and no such comparison is attempted here. The table-and-chair and storm costumes motivate the hypothesis; they do not test it.

Ambient $\mathbb{R}^3$ makes the expressions $\mathcal{S} + (a, b, c)$ and $\mathcal{S} \cdot (a, b, c)$ look legal. They are legal as *vector* arithmetic. They are not automatically legal as *unit* arithmetic. A state with $r = 0.98$ added to another with $r = 0.98$ along the same direction is $r = 1.96$, which has left the ball. A scalar $2 \cdot (0.98)$ does the same. The unit cap $r \in [0, 1]$ then forces a choice, to be made later and not here: clip (overflow saturates at 1), project (renormalize direction and forget extra strength), or refuse (the operator is undefined off the ball). Schoolbook $+$ on the `num` encoding is a different door: decode, add in $\mathbb{R}$,

encode back if the chart allows. Do not let $\mathbb{R}^3$ addition quietly replace $\oplus$, $\boxplus$, or T . The embedding lives in $\mathbb{R}^3$. The unit lives in the ball. As already noted in Section 5.8, the three-dimensional ball teaches the algebra; close document semantics, if pursued, live on a higher-dimensional sphere that carries the same verbs.

Open: dimensionality, $S^2 \rightarrow S^{N-1}$. Nothing stated above is intrinsically three-dimensional. The magnitude cap $r \in [0, 1]$ is a property of a vector’s length, not of the ambient dimension. The log and exponential maps used for the offset operator (Equations around (20)–(21)) are already Riemannian statements that hold on any $S^{N-1} \subset \mathbb{R}^N$, not only S^2 ; $\text{SO}(3)$ generalizes to $\text{SO}(N)$, and Axiom 2.8’s associativity and invertibility proofs are group-theoretic facts about $\text{SO}(N)$ for any N , not facts special to $N = 3$. Three dimensions were chosen for legibility and hand-checkable worked examples, not necessity. Raising N is expected to relieve one specific problem: too few directions forcing distinct concepts to crowd onto nearby points, a kind of clipping by dimensional scarcity rather than by magnitude. It does *not* relieve the separate overflow problem discussed just above (clip, project, or refuse when r exceeds 1) — that is a magnitude question and stays open regardless of N . This generalization is filed as a TODO, not claimed here: someone should restate the axioms for general N and check that every proof step (invertibility, associativity, the transport formula) survives verbatim rather than assuming it does.

Open: Mix across differing directions. Axiom 2.7 defines Mix only among states that share a direction $\hat{n}$, where it is phasor combination on χ . What Mix should mean for states at different directions is not settled here. An earlier draft of this paper carried a provisional recipe — normalize the vector sum $r_A \hat{n}_A + r_B \hat{n}_B$ — and it is withdrawn, because it is the only place in the construction where $\hat{n}$ would move by something other than a rotation. Everywhere else, direction is the exclusive property of $\text{SO}(3)$ (Axiom 2.8) and of T , and a blend that slides an arrow between axes is not a rotation. Three routes remain open, and this paper picks none of them: refuse Mix across directions outright, making the operator partial in place just as it is already partial in sort; rotate one state onto the other’s axis by an explicit, stated T first, so that all blending stays phasor blending; or accept a genuine two-stage operator and state separately what it preserves. Whichever is chosen, it should be shown that the resulting map still respects the cap and does not smuggle $\text{SO}(3)$ ’s job into $\boxplus$.

Open: reducing a path by merging close actors. Two *sequential* actors already compose into one equivalent rotation for free — that is just $\text{SO}(3)$ ’s (or $\text{SO}(N)$ ’s) closure, already latent in T ’s associativity. That is not the open question. The open question is whether several actors that are merely *close* — by some stated distance on $\mathcal{S}$, not identical — may be replaced by one representative actor before a path is walked, trading path length for approximation. This is a coarse-graining or clustering step, not composition, and it is not defined anywhere in this paper. Two things would need stating before it could be trusted: a distance and a merge rule (is the representative a $\boxplus$ -blend of the close actors, an average rotation, or something else), and a bound relating that distance to how far the reduced path’s endpoint may drift from the unreduced path’s endpoint. Composition is associative and exact; this proposed reduction is neither, and its error, if any, has not been characterized. A merge of this kind, whatever form it eventually takes, would produce an additional, derived actor for a shorter path to use — it would not

remove the original close actors from the bowl. The bowl under $\oplus$ (Section 2.2) already keeps every distinct seated unit; a reduced actor sits alongside its sources, not in place of them, and the unreduced path through the original actors remains walkable and exact. Reduction would be a second, optional reading of the same bowl, the way Mix and T already are, not a rewrite of what the bowl holds. Left open.